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Studierichting: Informatica
Oefeningenreeks 1c nr. 46,6

$$z^4 = 16$$

$$r = 16, \alpha = 0, n = 4$$

$$\rho = \sqrt[4]{16} = 2$$

$$\theta_k = \frac{0 + 2k\pi}{4}, k \in \{0, 1, 2, 3\}$$

$$k = 0$$

$$\theta_0 = \frac{0}{4}$$

$$z_0 = 2 \left(\cos \left(\frac{0}{4} \right) + i \sin \left(\frac{0}{4} \right) \right)$$

$$z_0 = 2$$

$$k = 1$$

$$\theta_1 = \frac{\pi}{4}$$

$$z_1 = 2 \left(\cos \left(\frac{\pi}{4} \right) + i \sin \left(\frac{\pi}{4} \right) \right)$$

$$z_1 = 2(0 + 1i) = 2i$$

$$k = 2$$

$$\theta_2 = \pi$$

$$z_2 = 2 \left(\cos \left(\frac{4\pi}{4} \right) + i \sin \left(\frac{4\pi}{4} \right) \right)$$

$$z_2 = 2(\cos(\pi) + i \sin(\pi))$$

$$z_2 = 2(-1 + 0i) = -2$$

$$k = 3$$

$$\theta_3 = \frac{\pi}{4}$$

$$z_3 = 2 \left(\cos \left(\frac{6\pi}{4} \right) + i \sin \left(\frac{6\pi}{4} \right) \right)$$

$$z_3 = 2(0 - 1i) = -2i$$

$$V \in \{2, 2i, -2, -2i\}$$

References